COMP4442

SERVICE AND CLOUD COMPUTING

Overview of
Service and Cloud Computing
Part-2

Applications of Cloud Computing

- Web applications
- Data backup,
- Disaster recovery,
- Office (email, virtual desktops, …)
- Software development and testing,
- Cloud applications in sectors
  - E-government,
  - Healthcare,
  - Education,
  - IoT,
  - Big data analytics,
  - Machine learning
  - …

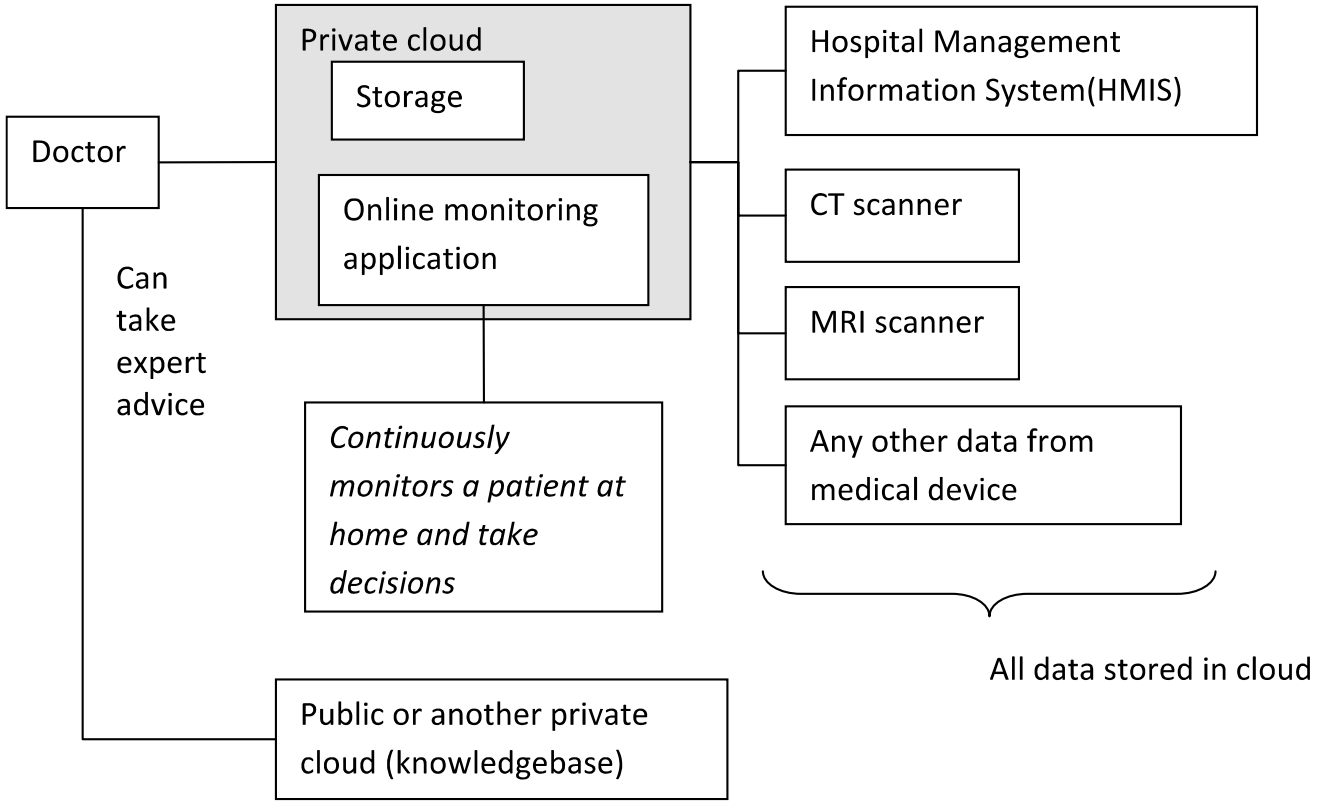

Fig. 1.20 Some of the roles cloud computing plays in healthcare

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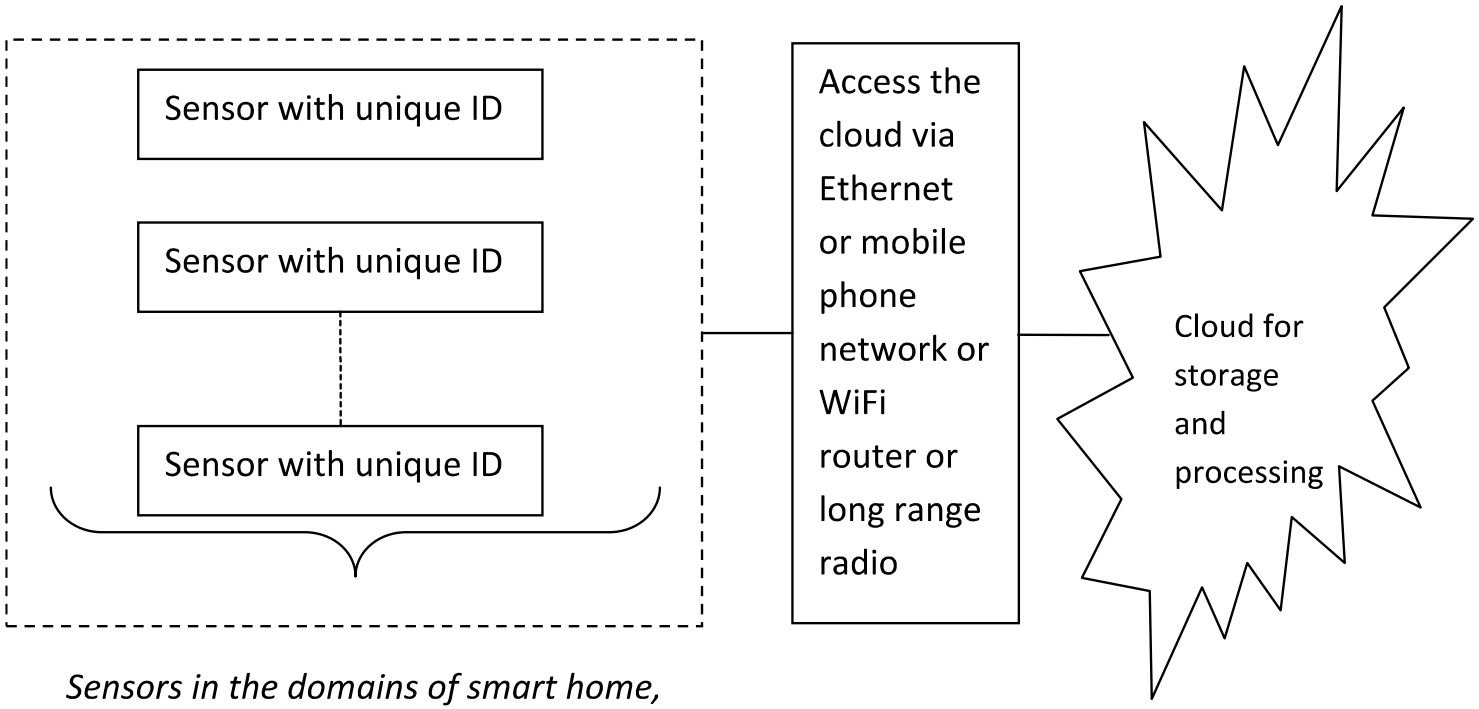


Sensors in the domains of smart home, smart healthcare, precision agriculture, weather, disaster detector data, etc.

Fig. 1.23 Cloud computing and the Internet of Things

Enabling Technologies of Cloud/Service Computing

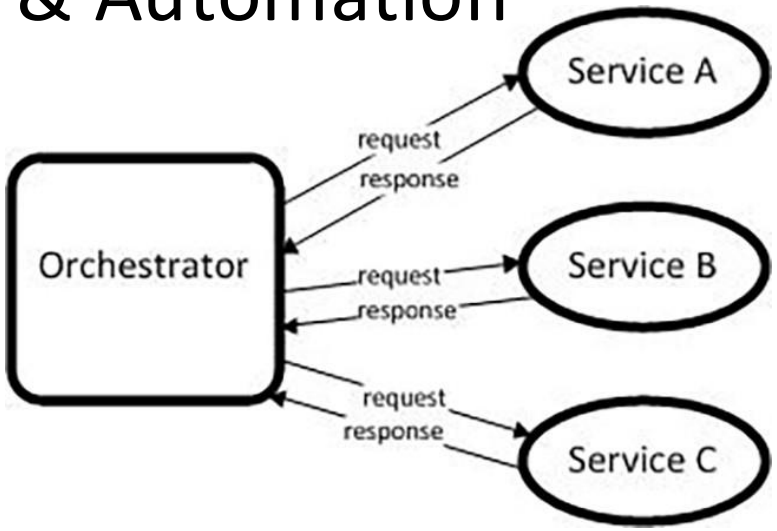
- Physical Infrastructure
  - Hardware resources: multi-core processors; server clusters; storage farm, networks, …
  - Data center facility
- Virtualization for multi-tenant resource sharing
  - Virtual machines: computing resources and storage resources
  - Containerization
  - Virtual networks, Software Defined Networks (SDN)
- Cloud computing platforms
  - Workload balancing
- Application architecture
  - Service Oriented Architecture (SOA), Micro-Services Architecture (MSA)
  - Web 2.0, PHP, Servlet, CGI
  - Spring+Tomcat, EJB, Java Persistence API (JPA), Java Database Connectivity (JDBC), Simple Object Access Protocol (SOAP), Representational State Transfer (REST)
- Cloud/service management
  - Functions for managing cloud/service systems

Cloud/Service Management: Service Request Management

- Provide a self-service interface for consumers to easily access various cloud services, e.g. service catalogues with SLAs (Service-Level-Agreement) and cost details.
- Either use *service portal* or *marketplace*
  - automation tools fulfil varying requests from users.
  - continuously updated with fresh features, functionalities, and facilities to gain an edge or retain the edge gained.
  - supported by operational team members employed by service providers

Cloud/Service Management: Orchestration & Automation

- Orchestration: compose different services from the cloud(s) into holistic workflow(s).



- Automation: automate cloud operations
  - Q&A What automation functions can you think of?

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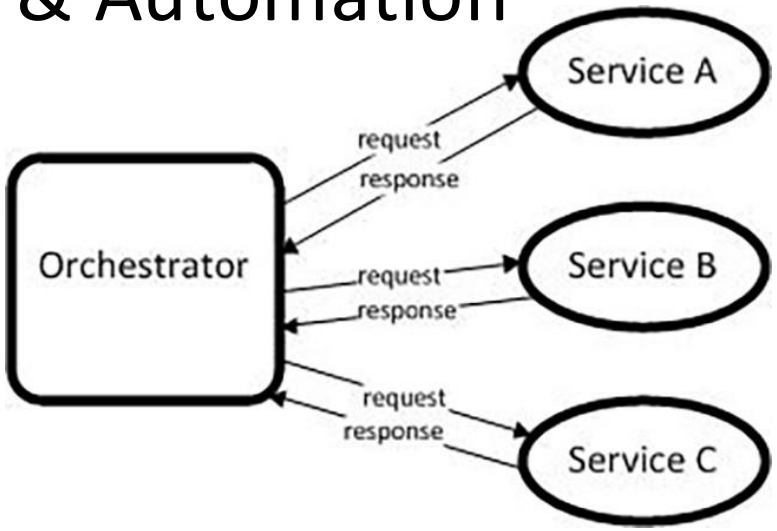


- Automation: automate cloud operations
  - Facilitate orchestration: infrastructure as code, wizard, brokerage
  - Provide version control for workflows
  - Scaling (up/down, vertical/horizontal) virtual servers
  - Load balancing
  - Backup data
  - Accounting and charging
  - Resource recycling

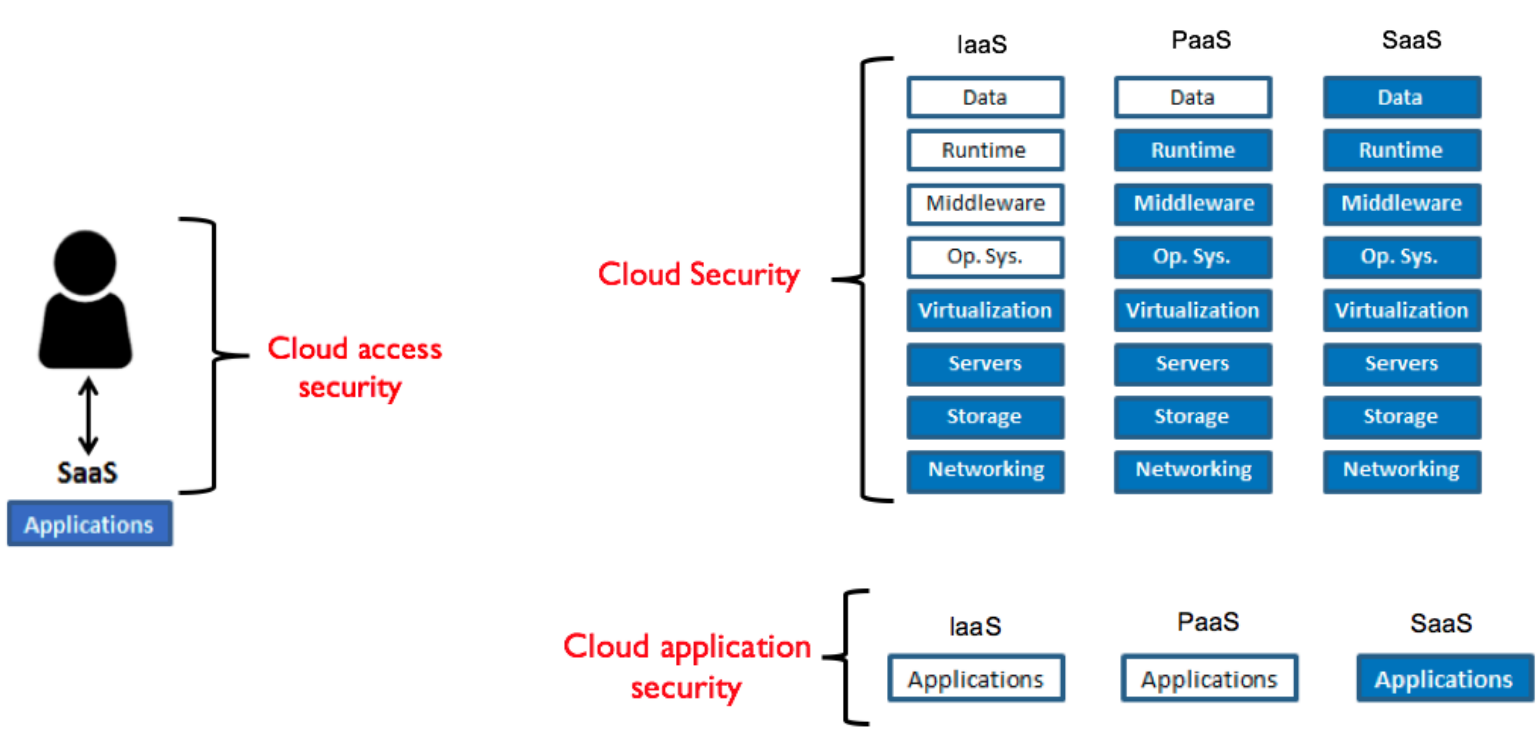
Cloud/Service Management: Service-Level Management

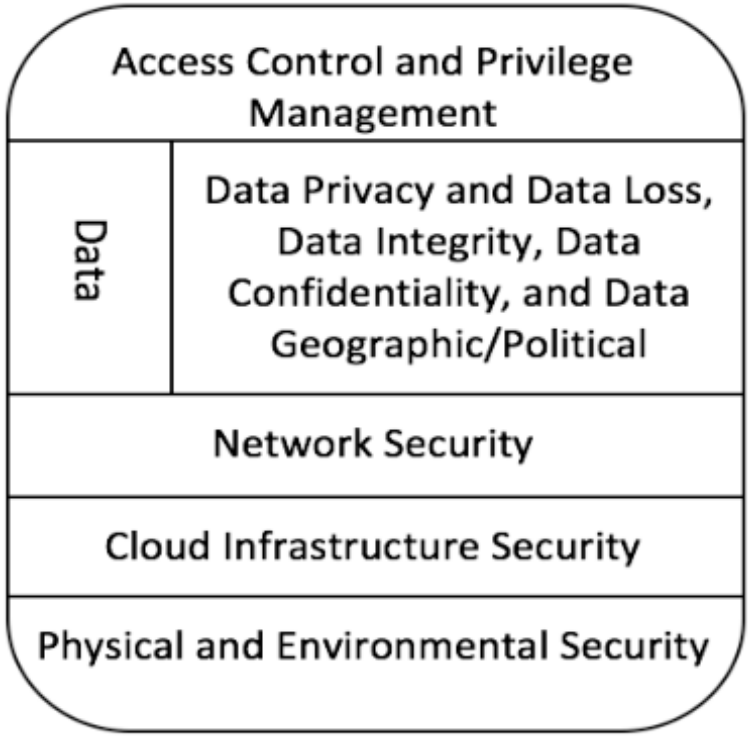
- Maintain the Service-Level Agreement (SLA) contracted between the cloud consumer and service.
- Often-requested SLAs:
  - Performance: response time, throughput, data/log quality.
  - Dependability: reliability, robustness, availability, fault prediction, fault tolerance, self-healing, security (confidentiality, authentication, assess control, integrity and immutability, denial-of-service, non-repudiation), privacy, forensics.
  - Others: scalability, flexibility, elasticity, prioritization, ease-of-use, ease-of-maintenance etc.
- How to monitor & maintain? (too big a problem space, following slides discuss just some of the topics)

Cloud/Service Management: Monitoring

- Constantly check the service-level
- In case of malfunctions or deviations from the SLA, appropriate troubleshooting should be carried out.
- Monitoring (including metering) in a cloud computing is much more challenging
  - Virtualized and distributed

Cloud/Service Management: Monitor/Maintain Security & Privacy





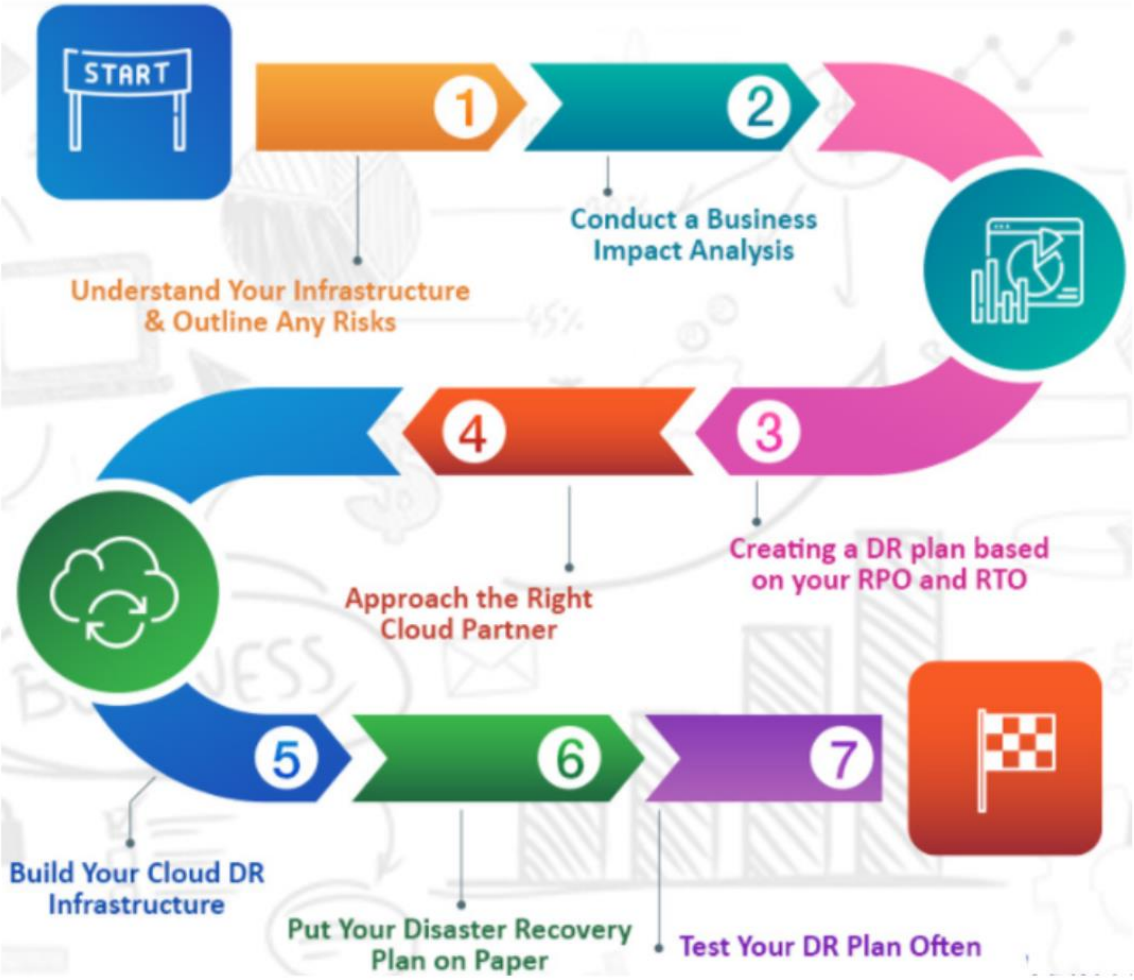
Q&A What cloud/service security/privacy challenges can you think of?

Cloud/Service Management: Monitor/Maintain Availability: Disaster Recovery (DR)

- Recovery Point Objective (RPO): e.g. guarantee no more than x-minute data loss.
- Recovery Time Objective (RTO): e.g. guarantee no more than x-minute down time.
- Q&A How?

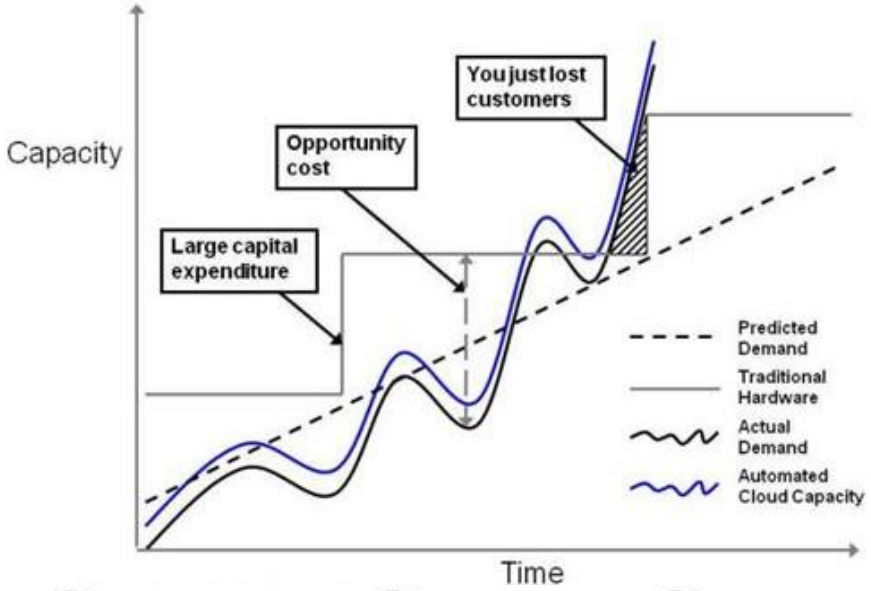
Cloud/Service Management: Monitor/Maintain Availability: Disaster Recovery (DR)

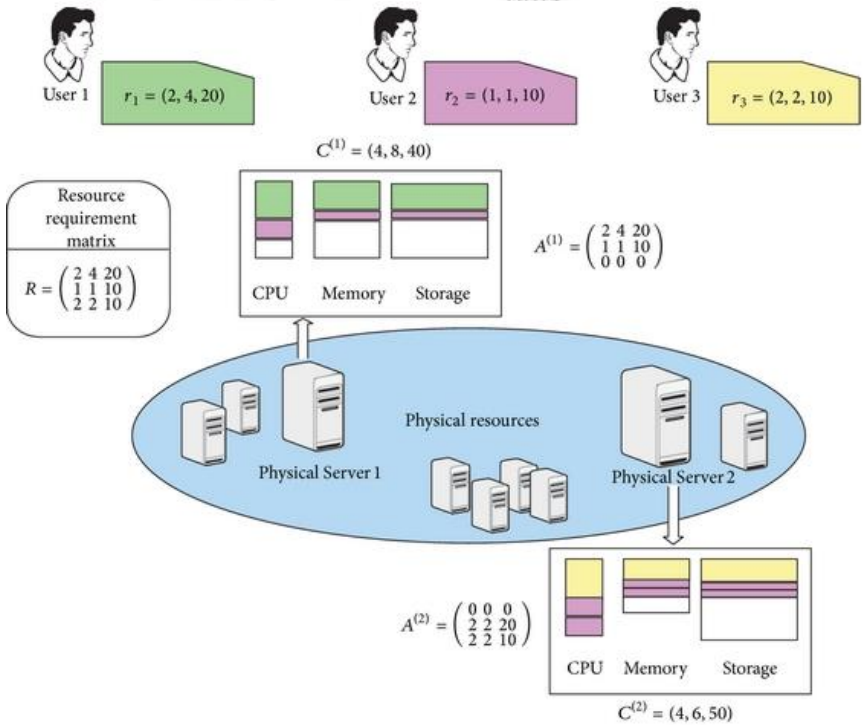
- Recovery Point Objective (RPO): e.g. guarantee no more than x-minute data loss.
- Recovery Time Objective (RTO): e.g. guarantee no more than x-minute down time.
- Have a second, independent cloud service provider as a backup solution to store copies of business-critical data.
- Traditional applications built in monolithic and legacy forms need to be consciously modernized for migration in cloud.



Cloud/Service Management: Monitor/Maintain Performance: Capacity and Resource Planning

- Dynamic capacity planning & resource allocation
  - Purposes:
    - dynamically accommodate computing capacity in order to meet SLAs
    - minimize the total costs incurred in the cloud services.
    - make efficient operational usage of the infrastructure footprint.
  - Often tied to orchestration and automation
  - Increasingly combined with cost transparency and optimization to enable tracking, budgeting, and optimization of cloud expenses





Open Source Tools for Building a Cloud System

- There are a number of solutions for creating a cloud architecture
  - Allowing us to compose an open source cloud framework (Fig. 6.1 in [b2]). However, a holistic solution is still lacking.

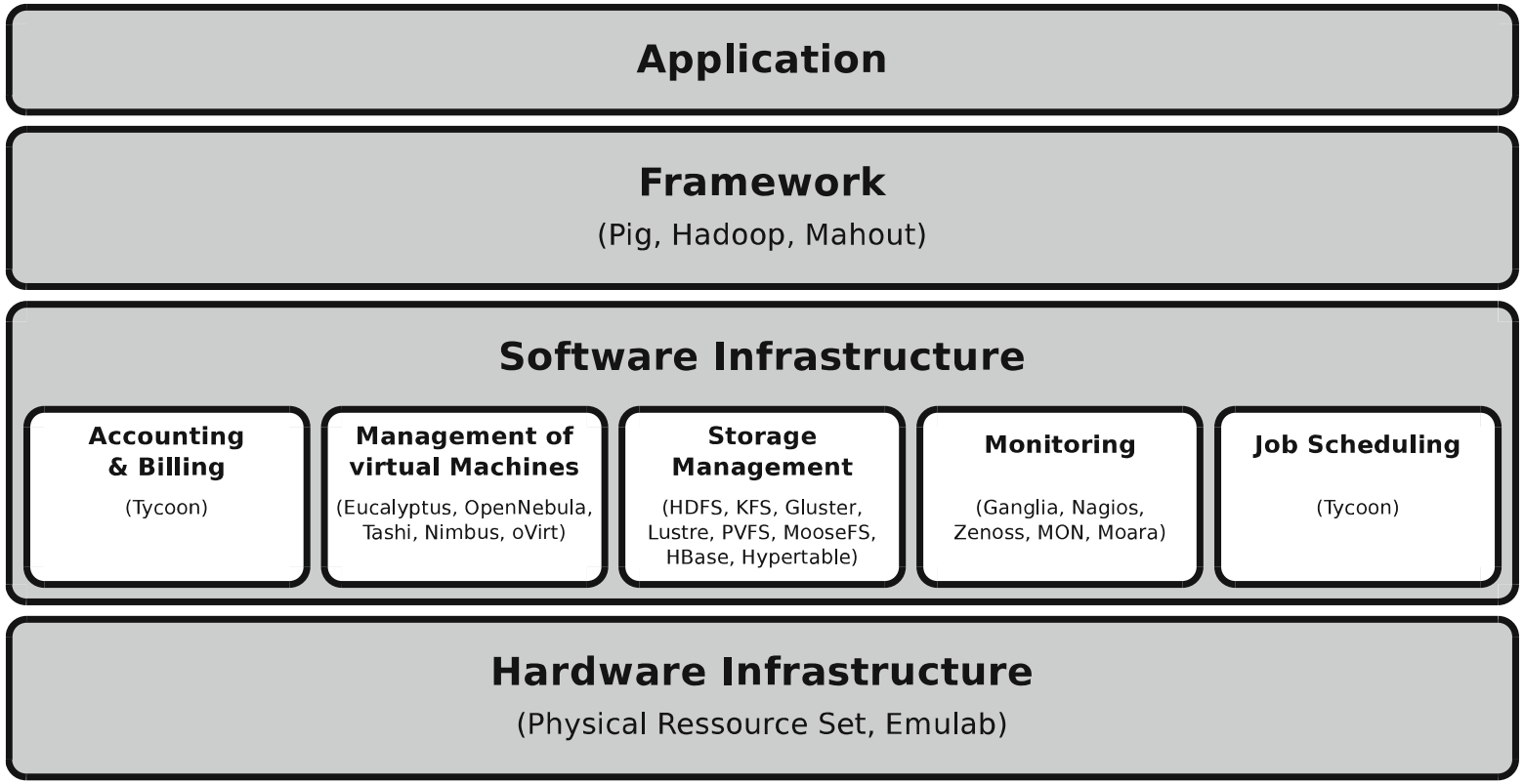


Fig. 6.1 OpenCirrus open source cloud stack [b2]

Open Source Tools for Build a Cloud System

- Physical and virtual resources
  - Infrastructure is organized into physical resource sets (PRS)
    - Resources in each PRS are linked by a virtual LAN in a common domain
    - Domain management: a special service used to manage resources over the network and to allow to switch components on or off, roll out system images, and monitor the infrastructure
  - Virtual clusters form virtual resource sets (VRS) which can be managed by systems like Eucalyptus and Tashi.
    - Eucalyptus allows to set up and operate an independent IaaS cloud infrastructure, with API compatible with Amazon EC2, S3, and EBS
    - Other IaaS solutions include OpenNebula, Nimbus, CloudStack, OpenStack

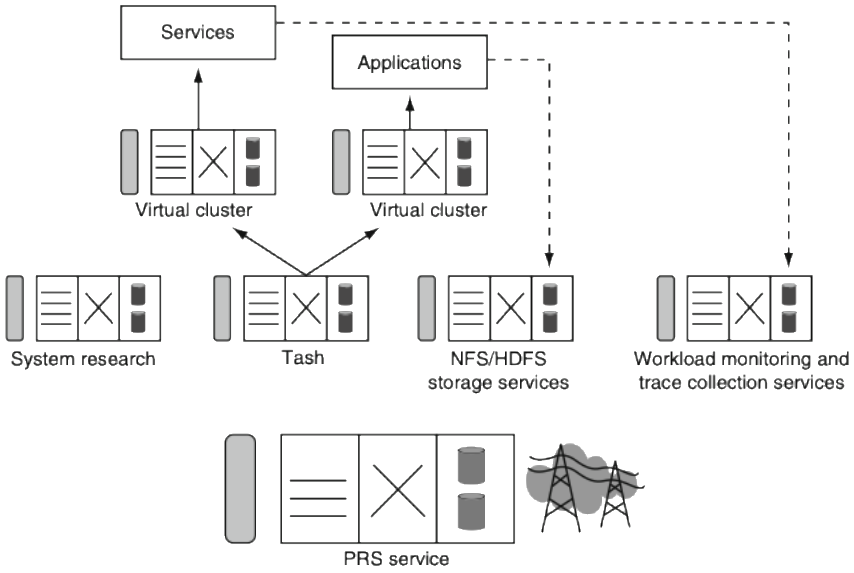


Fig. 6.2 Different domains associated with a physical resource set

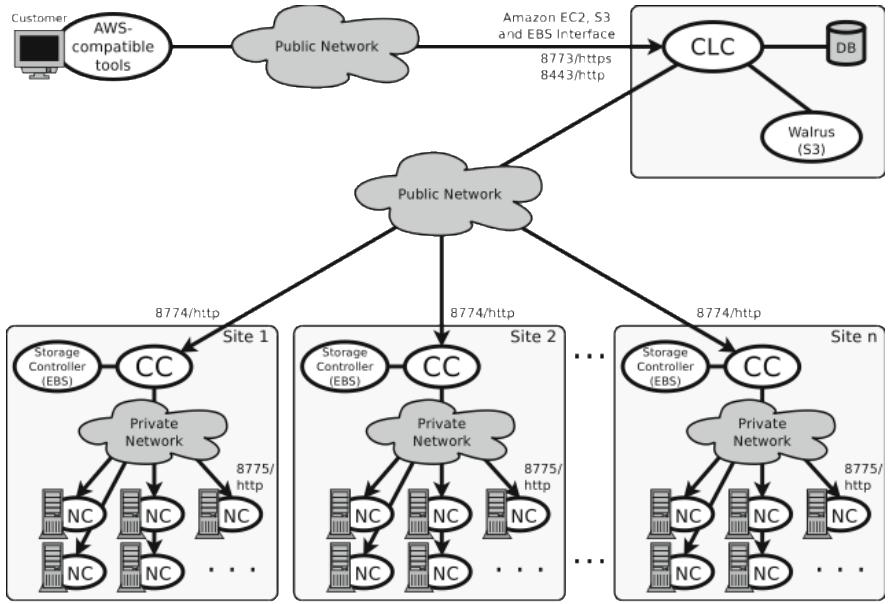


Fig. 6.3 Eucalyptus architecture and components

Putting All Together

Three major components:

1. Frontend: User
2. Network: Access
3. Backend: Cloud

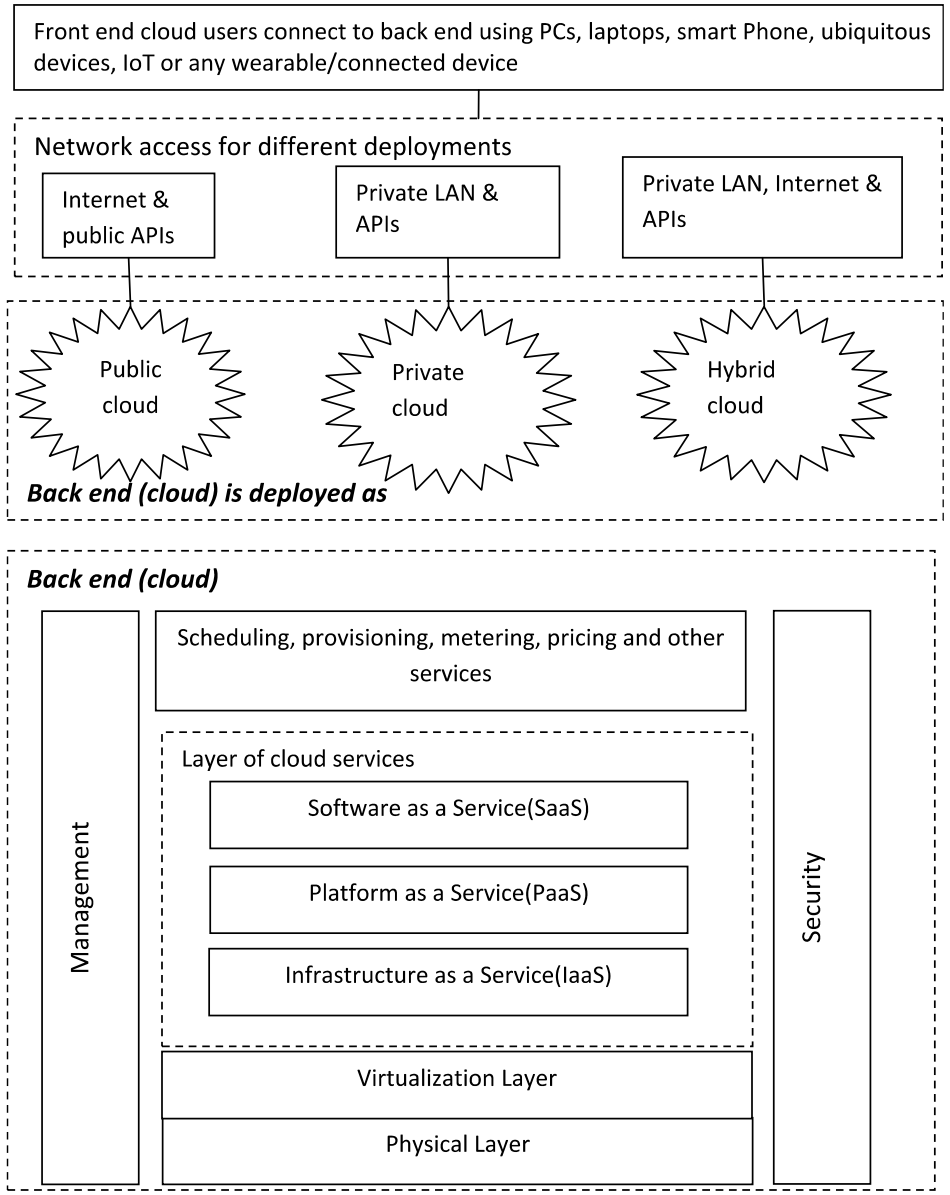


Fig. 2.6 Cloud architecture